Housing First or Treatment First?

A stylized model, and the quantitative model behind it

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1 A stylized model

1.1 Three questions

In the data, drug use is concentrated among low income households, most of the homeless do not carry a substance use disorder, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has six state variables and roughly forty parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the work. This note strips the environment to the smallest one in which my three main questions are still well posed.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution. Drugs are a normal good bought at a price, so that concentration is not what a model produces by default. Which feature of the environment reverses the concentration from the top to the bottom?
2. **How does housing assistance shape the drug use decision?** A voucher pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: call this the value of life force, and it lowers use. In contrast, it cushions the consequence of acquiring the disorder, because a household that acquires one is housed anyway: call this the moral hazard force, and it raises use. Both are first order, so theory does not sign the effect of housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First better than Housing First?** In a model without externalities, never. Housing First always raises welfare when agents are forward looking, as they are here. Addiction creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension alone cannot justify conditionality. However, if addiction and homelessness generate an externality, such as reducing the earning potential of children, then Treatment First may be preferred. The model's work here is to show why the first claim holds and then to qualitatively price the externality that overturns it.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and every component that obscures the mechanisms has been removed and the model refitted without it. What remains is what the results need. The welfare evaluations below are for illustration only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

1.2 The environment

1.2.1 Households and preferences

There are three periods and a continuum of households. A period stands for twenty years of adult life, so the three of them span ages twenty five to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is 0.99 annually and $\beta = 0.99^{20} = 0.818$ per period, the gross interest rate is $R = 1/\beta = 1.223$ so that $\beta R = 1$, and ϕ_0 is survival over twenty years. Notation follows the quantitative model throughout.

A household is characterised by its permanent productivity y, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma^2, \sigma^2)$, so mean income is one. Households differ in y, in age $j \in \{1, 2, 3\}$, in employment status $e \in \{0, 1\}$, in voucher receipt $\eta \in \{0, 1\}$ and in disorder status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility depends on non durable consumption c, housing services h and drug consumption v,

$$u(c,h,v) \;=\; (1-\gamma-\theta)\log c \;+\; \gamma\log h \;+\; \theta\big[\log(b+v)-\log b\big] \;+\; \aleph. \tag{1}$$

The three weights sum to one, so γ is the housing budget share proper. Drugs enter only in the first period, since that is the only period in which v is chosen, and housing is chosen each period from a three point grid $\{h_0, h_S, h_L\}$. The pair (θ, b) governs the drug term: the marginal utility of the vth unit is $\theta/(b+v)$, so θ mostly sets how much a user uses and b whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, since under plain $\theta\log v$ every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish when consumption is low. Households save in one risk free asset and cannot borrow.

1.2.2 Timing

Period 1. Every household is employed and clean. It chooses consumption, housing, drug use $v \geq 0$, which for simplicity is the only period in which it chooses whether to use, and saving, paying market rent p_r, which is normalised to one, on whatever unit it picks. Using drugs generates the risk of a substance use disorder next period, increasing in v,

$$\pi(v) \;=\; 1-e^{-\lambda v}. \tag{2}$$

Because π is increasing in v, the household internalises that using more is riskier, so the intensive and extensive margins are not separate decisions.

Period 2. For households that used, a disorder $d \in \{0,1\}$ is realised with probability $\pi(v)$; for the rest no disorder is realised. A household without the disorder is unemployed with probability u_0, and a household with the disorder is unemployed with the higher probability u_1. If unemployed, it receives income support. The household then chooses consumption, housing and saving. The probability of recovering from the disorder by the next period is ρ.

Period 3. Terminal, after which the household dies with certainty. Employment, income support, the voucher and the housing choice work as in period 2, there is no drug use decision, and assets are run down with nothing bequeathed.

Mortality, and why there are three periods. A household survives from one period to the next with probability

$$\phi(d) \;=\; \phi_0\,(1-\kappa d), \tag{3}$$

so κ is the proportional reduction in survival caused by the disorder. The third period is what makes that reduction bite. Mortality is a tax on a continuation value, so it can only matter if there is a period after the one in which the disorder is carried. In a two period version the disorder is realised in the last period, death costs nothing because nothing follows it, and the mechanism the whole model rests on is silent. Three periods is the smallest number at which acquiring the disorder has a future to destroy.

1.2.3 Employment, income support and housing

The gap $u_1 - u_0$ is the entire resource cost of the disorder. Income in a period, given permanent productivity y and employment status e, is

$$y^g(y,e) \;=\; \begin{cases} y & \text{if } e=1, \\ \psi\cdot\min\{y,\bar{y}\} & \text{if } e=0, \end{cases} \tag{4}$$

with ψ the replacement rate and $\bar{y}$ a cap on the earnings base. There is no other income risk. The level of ψ is what links the disorder to homelessness: for the poorest households half of income can fall short of the rent on the smallest unit, so an unemployed household there cannot house itself out of current income.

Each period the household chooses

$$h \in \{h_0,\ h_S,\ h_L\}, \qquad h_0 < h_S < h_L, \tag{5}$$

homelessness, a small unit and a standard unit. Homelessness is free and delivers almost nothing. Two rented sizes is the fewest that lets a household economise on housing rather than lose it: with one, homelessness comes out at 3.48 per cent against a target of 0.40, at any h_0.

A voucher covers part of the rent, is means tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0, 1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \ = \ \pi_V \cdot \mathbf{1}\{y' \le y_V\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{6}$$

and a household holding one pays the fraction κ_r of market rent, so the rent paid on option j is

$$r_j(\eta) \ = \ p_r\, h_j \big[\kappa_r \eta + (1 - \eta)\big] \cdot \mathbf{1}\{j \neq 0\}, \tag{7}$$

which is zero when the household is homeless. Vouchers operate in periods 2 and 3, and π_V, the share of eligible households actually covered, is the policy dial of Section 1.5.

The two policies differ in exactly one respect, the last indicator in equation (6). **Housing First** covers every eligible household regardless of (d', v). **Treatment First** covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. Since cover in the states that follow drug use is the only difference, the gap between them is the moral hazard and the whole of it.

1.2.4 The household's problem

Let a be assets carried in. In period 3 the household consumes everything and chooses housing,

$$W_3(a, y, e, \eta) = \max_j \Big\{ (1 - \gamma - \theta) \log c + \gamma \log h_j + \aleph \Big\}, \qquad c = Ra + y^g(y, e) - r_j(\eta). \tag{8}$$

In period 2 it also saves, and faces mortality and recovery,

$$W_2(a, y, e, \eta, d) = \max_{j,\, a' \ge 0} \Big\{ (1 - \gamma - \theta) \log c + \gamma \log h_j + \aleph + \beta\, \phi(d)\, \mathbb{E}\, W_3(a', y, e', \eta') \Big\}, \tag{9}$$

with $c = Ra + y^g(y, e) - r_j(\eta) - a'$, and the expectation taken over next period's employment and voucher draws, which depend on the disorder status d', itself equal to zero with probability ρ when $d = 1$ and to d otherwise. In period 1 the household is employed, clean and holds no voucher, so its state is y alone, and it chooses housing, drug use and saving,

$$V_1(y) = \max_{j,\, v \ge 0,\, a' \ge 0} \Big\{ (1 - \gamma - \theta) \log c + \gamma \log h_j + \theta[\log(b + v) - \log b] + \aleph + \beta\, \phi_0\, \mathbb{E}\, W_2(a', y, e', \eta', d') \Big\}, \tag{10}$$

with $c = y - r_j(0) - p_v v - a'$ and $d' = 1$ with the use dependent probability of (2). A household uses if the value at its best positive v exceeds the value at $v = 0$, and because households differ only in y that defines a set of income intervals on which use occurs.

1.3 Calibration

Table 1 lists the parameters set outside the model. Preferences and demographics are conventional: the annual discount factor is 0.99 and the gross interest rate is $1/\beta$, the dispersion of log permanent income is $\sigma = 0.75$ from Heathcote, Perri and Violante (2010), and per period survival is $\phi_0 = 0.93$ from the CDC life tables. For the disorder, recovery over a period is $\rho = 0.40$ from the SAMHSA treatment episode data, the proportional fall in survival is $\kappa = 0.45$ from Hjorthøj et al. (2023), and the two unemployment probabilities are measured directly, $u_0 = 0.049$ from the BLS and $u_1 = 0.1625$ from NSDUH 2024 for respondents aged 26

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor over a twenty year period	0.99 annually, compounded	0.818
R	Gross interest rate over a twenty year period	Set to $1/\beta$	1.223
σ	Dispersion of log permanent income	Heathcote, Perri and Violante (2010)	0.75
ϕ_0	Survival per period, no disorder	CDC life tables	0.93
θ	Weight on drug use	Set; see the text	0.11
γ	Weight on housing services	$0.25(1-\theta)$, Consumer Expenditure Survey	0.223
The disorder			
ρ	Recovery probability per period	SAMHSA, TEDS	0.40
κ	Proportional fall in survival with the disorder	Hjorthøj et al. (2023)	0.45
u_0	Unemployment probability, no disorder	BLS	0.049
u_1	Unemployment probability, with the disorder	NSDUH 2024, ages 26 and over, weighted	0.1625
Income support			
ψ	Replacement rate while unemployed	DOL, statutory UI rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	DOL, state benefit maxima	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Normalisation	1.00
p_v	Price of a unit of drugs	Normalisation of the units of v	1.00
h_L	Standard rental unit	ACS 2023, rent over income	0.212
h_S	Small rental unit	Half the standard unit	0.106
κ_r	Tenant's share of the rent when covered	HUD voucher rules	0.30
y_V	Voucher means test, multiple of average income	HUD income limits	1.00
π_V	Voucher generosity in the baseline	HUD, share of eligible assisted	0.27

and over. Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules. The standard rental unit costs 21.2 per cent of mean income, its rent to income ratio in the ACS, and the small unit is half of it. The voucher parameters are institutional: covered tenants pay 30 per cent of market rent, eligibility is capped at mean income, and $\pi_V = 0.27$ is the HUD share of eligible households actually assisted.

Four parameters are set inside the model, in Table 2, each bracketed on the moment it moves most rather than fitted jointly: with no preference heterogeneity the aggregate is a cliff in b, so a gradient method steps out of the region where participation is interior and walks to a corner. The flow value of being alive $\aleph$ is matched to a value of a statistical life of 80 times average annual income, about \$9 million against a mean household income of \$113,000, which is the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit b is matched to a drug participation rate of 5.13 per cent, housing services on the street h_0 to a homelessness rate of 0.399 per cent from the HUD point in time counts, and the addiction hazard λ to the 33.2 per cent of users who acquire a disorder.

The weight on drugs θ is set rather than calibrated, and the reason is worth stating because the obvious alternative was tried. The natural target is the 11 per cent of income that users who are not addicted spend on drugs, which the quantitative model matches, and this model cannot reach it while keeping users poor. Two moments already pin the pair (λ, v): the disorder rate among users, $1 - e^{-\lambda v} = 0.332$, fixes λv at 0.403, and an 11 per cent spending share on an average user income near 0.2 would fix v near 0.022 and so λ near 18. Scanning (λ, θ) over that region shows the model is bimodal with no interior solution: below a threshold in θ users are the very poor and spend a hair above nothing, above it users are the rich and the income gradient the model exists to explain is gone. The economics is not subtle. What puts use at the bottom is that the disorder destroys a continuation value, and for a poor household that cost is enormous relative to the utility of the drug, so it uses as little as it can while still using; low intensity use among the poor is the same fact as the income gradient here, not an independent one. Setting $\theta = 0.11$ and $\gamma = 0.25(1-\theta) = 0.223$, the model produces a spending share of 1.76 per cent. It misses that moment by a wide margin, and the quantitative model, with its separate light and heavy experimentation margins and its drug demand from households that already carry the disorder, is where it should be matched.

Table 2: Internally calibrated parameters

Parameter	Description	Target moment	Data	Model	Value
$\aleph$	Flow value of being alive	Value of a statistical life, annual incomes	80.0	80.0	2.855
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00032
h_0	Housing services when homeless	Homelessness rate	0.399%	0.371%	0.00073
λ	Addiction hazard per unit of use	Share of users who acquire the disorder	33.2%	36.3%	783.1

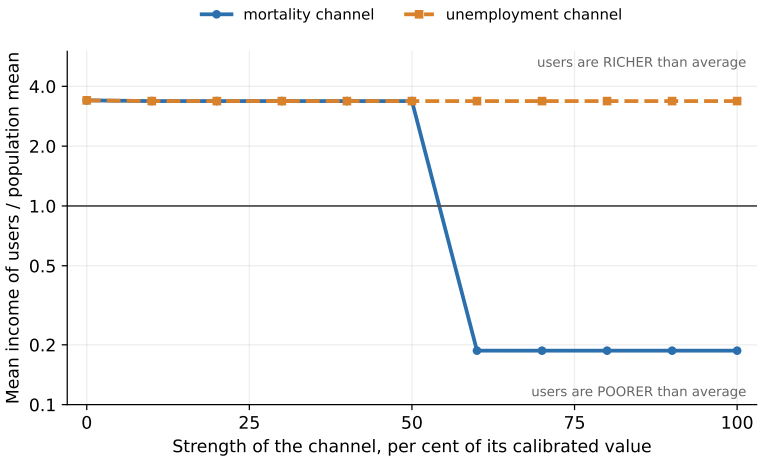


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is average income of users over the population average, on a log scale, and b is recalibrated at every point so that participation is 5.13 per cent throughout.

1.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a disorder that is the only force operating, and drug use concentrates at the top: the average user then earns 3.40 times the population average.

Two features of the environment could reverse that. The *mortality channel* is that the disorder kills you, through κ in equation (3). The *unemployment channel* is that it costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize, live on less, or lose your home. Figure 1 scales each from nothing to its calibrated value, recalibrating b at every point so participation stays at 5.13 per cent. The amount of drug use is therefore identical all along both curves and the only thing moving is which households use.

The result is sharper than the statement that mortality matters more. That channel reverses the gradient at half of its calibrated strength, taking the income ratio from 3.40 to 0.19, while the unemployment channel does not reverse it at all, leaving users at 3.37 times average income even fully switched on. A resource penalty cannot produce the concentration of drug use at the bottom.

A mortality penalty is different in kind. A resource penalty scales with the very thing it is meant to offset, since taking a fixed share of a large income still leaves a large income, so it cannot outrun the fact that drugs are cheaper in utility terms for the rich. What mortality destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and loses proportionally more than it gains from drugs being relatively cheap. That is the work of $\aleph$ in equation (1): it is the part of the continuation value that does not shrink as consumption falls.

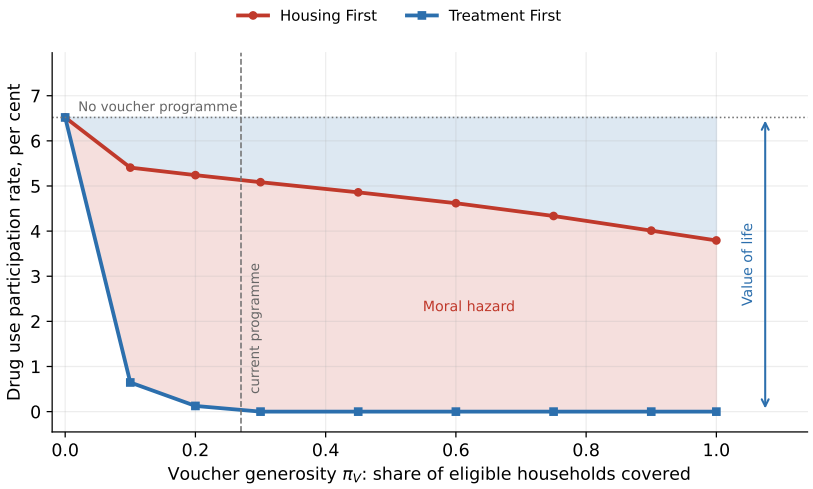


Figure 2: Drug use under the two policies as the voucher becomes more generous

Note: the red band is the moral hazard force and the blue band above it is what Housing First delivers net; together they are the value of life force, marked at the right. The dashed line is the generosity of the current programme, $\pi_V = 0.27$.

1.5 How does housing assistance shape the drug use decision?

The two forces of question 2 can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value of life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value of life force.

Figure 2 traces both as generosity rises. With no programme at all, 6.52 per cent of households use. Universal Treatment First takes that to 0.00 per cent, a value of life force of -6.52 percentage points; universal Housing First leaves it at 3.79 per cent, so the moral hazard force is +3.79 points and the net effect of Housing First is -2.73. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

Figure 3 plots the homelessness and disorder outcomes for each programme as its generosity changes, alongside drug use. On homelessness the two policies end in the same place, both driving it from 0.78 per cent with no programme to zero, and along the way Treatment First is if anything slightly below Housing First, because the households it deters from using never acquire the disorder that puts their housing at risk. The larger difference is on the other margin, where at universal cover Housing First leaves 0.602 per cent of the population carrying the disorder and Treatment First leaves 0.000 per cent. What the moral hazard buys is extra disorder cases among households that are housed either way.

1.6 Welfare

1.6.1 Without externalities, Housing First always wins

Sections 1.4 and 1.5 establish what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity π_V, whose outcomes are in Panel A of Table 3. The policies differ only in the last indicator in equation (6), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a household that optimises cannot be worse off:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{11}$$

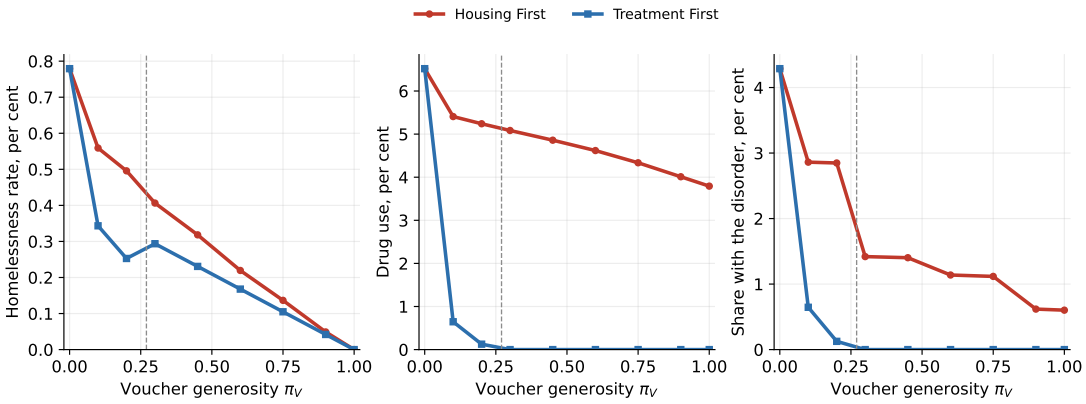


Figure 3: Homelessness, drug use and the disorder under the two policies

Table 3: Welfare at universal cover, with no externality

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	3.79%	0.00%
Share with the disorder	0.602%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.1694	0.1696
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		4.1%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.41%
Consumption equivalent loss, whole population		0.017%

Note: the consumption equivalent is the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value. Voucher outlay is expected spending on a household from period 2 on, discounted, per unit of average income.

with equality exactly for households that never use. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.[1]

It follows that the moral hazard is not a distortion. Households here face the true addiction hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is now better able to bear. The +3.79 points of Section 1.5 are a statement about drug use and not about welfare. Panel B of Table 3 puts numbers on it: 4.1 per cent of households are strictly worse off under Treatment First, none is better off, and those affected lose 0.41 per cent of consumption.

A fixed budget does not rescue conditionality either. Treatment First is actually slightly *more* expensive at universal cover, 0.1696 against 0.1694, so matching on cost rather than on generosity changes nothing. The reason is that survival to period 3 falls from 0.93 to 0.51 with the disorder, so the households the moral hazard adds to the rolls die sooner and draw voucher payments for fewer periods. So a case for conditionality has to come from an externality.

1.6.2 The externality on children reverses it

The externality I model is one that runs through income rather than through preferences. I have considered two alternatives, but both seem to be at odds with reality. Letting the child of a user inherit a stronger taste,

[1] The same comparison in the quantitative model gives the same answer.

Table 4: The externality on children

Panel A. Transmission: what each income penalty implies			
δ_d	Child's disorder rate	Ratio	Loss to the child
0.00	2.20%	1.00	0.0%
0.20	3.95%	1.80	21.7%
0.30 (calibrated)	5.34%	2.43	37.3%
0.40	7.27%	3.31	58.2%
Panel B. Does the externality reverse the ranking?			
Private welfare loss from conditionality			0.017%
Disorder cases prevented			0.602 pp
Damage to the child of one such case			34%
Externality gain from conditionality			0.202%
Gain over loss			12.1

Note: Panel A is at the baseline programme, $\pi_V = 0.27$, since the transmission ratio in the data comes from an economy with a rationed programme; Panel B is at universal cover, one child per household, no discounting across generations. Losses are consumption equivalents.

a lower b, matches the transmission fact but has the wrong welfare sign: b enters only through $\theta \log(1 + v/b)$, so a lower one raises the child's objective at every $v > 0$ and its lifetime value with it. Letting it inherit a greater susceptibility, a higher λ, fails the other way, since a child facing a riskier drug optimally uses *less* and is worse off ex ante, which is clearly counterintuitive.

Income works on both counts, and it works through the channels explored in Section 1.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. The income gradient does the whole job, so the model transmits addiction across generations *without transmitting any preference for drugs.*

A child's baseline income is drawn from the same lognormal as everyone else, and the entire link is the penalty

$$y_{\text{child}} \;=\; y_{\text{base}}\,(1-\delta_d)^{\mathbf{1}\{\text{parent acquired the disorder}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}, \tag{12}$$

which the parent does not internalise. That is what makes it an externality rather than something already priced into V_1.

The two penalties are set to different things. Panel A of Table 4 shows that $\delta_d = 0.30$ makes the child of a parent who acquired the disorder 2.4 times as likely to acquire one, in the middle of the epidemiological range of two to four in McGovern et al. (2023) and the alcohol literature. The homelessness penalty δ_h is set directly from Chetty, Hendren and Katz (2016), who find that children who moved to a low poverty neighbourhood before age 13 went on to earn 31 per cent more as adults. The two penalties also do different work: both policies drive homelessness to zero at universal cover, so δ_h prices the programme against no programme while δ_d decides Treatment First against Housing First.

Panel B puts the two sides together and they are not close. The private cost of conditionality, in the first row, is 0.017 per cent of consumption, small because only 4.1 per cent of households are affected. Against that, conditionality prevents 0.602 percentage points of disorder cases in the second row, and each of those cases costs one child 33.6 per cent of consumption, in the third. The externality gain is 12.1 times the private loss, and stays above it throughout the range of δ_d in Panel A. Housing First is better for every household taken one at a time, and worse once one counts the children of the households it fails to deter.

Treatment First drives drug use to *exactly* zero here, and that inflates the margin. Nobody in this model faces the sobriety condition and fails to meet it, and the households who cannot comply are the entire empirical case for Housing First.

The point of this section is only that an externality can reverse a ranking that is otherwise airtight, and one externality is enough to show it. Which way the comparison actually goes is a question about magnitudes, and that is what the quantitative model is for.

2 The quantitative model

This section states the model the paper's numbers come from, completely. Where the stylized model already carries something unchanged it is named rather than motivated a second time: the Stone-Geary form of the drug term, the logic that separates the two policies, and the transmission externality all appear below exactly as Sections 1.2 to 1.6.2 give them. Everything else is written out, because it is what turns a directional answer into a quantitative one.

2.1 Demographics, the state and the choice set

Households enter at age 26, live at most to 85, so $j = 1, \ldots, J = 60$ annual periods, and retire at $j_R = 41$. Survival from j to $j+1$ is $\phi_j(h', d, m)$ in (27): it depends on age, on the housing *chosen* this period, on the disorder and on mental illness. Newborns replace the dead. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathcal{A} \times \mathcal{Y} \times \{1,2,3\} \times \{0,1\}^3, \tag{13}$$

assets $a \geq 0$ on a 60 point grid dense near zero, permanent productivity y on 9 points, employment e, the disorder d, serious mental illness m and the voucher η, which is 12,960 points per age and 777,600 in total.

Two economies in that list are choices rather than oversights. Employment takes three values, employed, unemployed with insurance and unemployed without, because insurance is drawn on entry into a spell and lasts one year, so a first year non recipient and a household unemployed for two years or more have identical income and identical continuation values and merge *exactly*. And the housing unit a household arrives in is not a state: everything that conditions on housing conditions on the unit chosen, and renters face no adjustment cost, so carrying it would replicate the whole problem four times over for nothing. It would have to be reinstated if moving costs were added.

Each period the household chooses

$$\big(n,\, h',\, v,\, \tau,\, a'\big) \in [0,1] \times \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A}, \tag{14}$$

hours n, housing h' where h_1 is homelessness, drug use v, treatment take up τ, available only when $d = 1$, and next period assets a'. Every shock is drawn *after* all of that, which is what makes those variables state variables rather than choices, and it is why the transitions in Section 2.10 condition on the chosen h', v and τ: the choice comes first, the draw second.

2.2 Preferences

Period utility is

$$u\big(c, h', v, n, \tau; d, m\big) \;=\; \frac{\big[\, c^{\alpha}\,(h')^{\beta}\,(v + b_{dmh'})^{\gamma}\,\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\bar{\tau}^{u} \;-\; \kappa_x\big(H + \omega_x H_D\big), \tag{15}$$

with $\alpha + \beta + \gamma = 1$. Equation (1) is the logarithmic case of the bundle with hours fixed, so γ and b do what θ and b do in Section 1.2: b is the Stone-Geary endowment that keeps the marginal utility of the first unit finite and therefore decides whether anyone uses at all.

Four things are new. Leisure makes hours a margin. Treatment costs utility rather than money, which is what the evidence supports, since the three largest barriers reported in the NSDUH are beliefs and preferences and not prices. The term $\kappa_x(H + \omega_x H_D)$ prices the disamenity of the aggregate homelessness rate H, weighted by the share H_D carrying the disorder; it is zero in the benchmark, for the reason Section 2.16 gives. And the Stone-Geary shifter itself varies,

$$b_{dmh'} = b_d \cdot \rho_m^{\mathbf{1}\{m=1\}} \cdot \rho_H^{\mathbf{1}\{h'=h_1\}}, \qquad \rho_m, \rho_H \in (0,1], \tag{16}$$

so the disorder, mental illness and homelessness each raise the marginal utility of the first unit of the drug. The Cobb-Douglas weights themselves are held fixed across the disorder: only the shifter moves. This matters for what follows, because $\rho_H < 1$ is one of the four channels through which homelessness and the disorder reinforce each other.

2.3 Income, taxes and transfers

Gross income depends on age, employment, productivity and health,

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq j_R \quad \text{(social security)} \\ n\,y\,\varepsilon_j\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j<j_R,\ e=1 \quad \text{(employed)} \\ \varrho_{UI}\,y\,\varepsilon_j & j<j_R,\ e=2 \quad \text{(insured, first year)} \\ \zeta_{DI}\,\bar{y}_{DI}\,\bar{Y} & j<j_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{17}$$

with ε_j the deterministic age profile of efficiency, ρ_d and ρ_m productivity penalties applied *conditional on working*, ϱ_{UI} the replacement rate paid on the household's own productivity, ζ_{DI} the share of the non employed with mental illness who receive disability, and $\bar{Y}$ average income. Retirement income replaces 40 per cent of the productivity state with no health penalty, since social security is not means tested on health.

The conditioning of the penalties on *working* is what lets an employed household be poor enough to choose homelessness. An unemployment effect alone cannot do that, and the data require it: the homeless are roughly half in work, not none. Section 1.2.3's environment, where only unemployment lowers income, is exactly the version that cannot.

Labour income is taxed with the Heathcote, Storesletten and Violante function and capital income at a flat rate,

$$y^n = \tau_\ell\,(y^g)^{\xi} \;+\; (1-\tau_k)\,r\,a, \tag{18}$$

where ξ governs progressivity and τ_ℓ the level, and τ_ℓ is what clears the government budget. Two transfers follow, both means tested on gross income as the programmes themselves are. A food stamp floor tops means tested income up to $\underline{c}\bar{Y}$, and the voucher covers rent above a tenant contribution of κ_r of gross earnings,

$$F = \max\big\{0,\ \underline{c}\,\bar{Y} - (y^g + \max\{0, ra\})\big\}, \qquad Vch = \mathbf{1}\{\eta=1\}\,\mathbf{1}\{h'\neq h_1\}\,\max\big\{0,\ p_h h' - \kappa_r\,y^g\big\}. \tag{19}$$

A homeless voucher holder does not lose the voucher: retention depends on income eligibility, not on housing, so the entitlement survives and simply pays nothing until the household rents again.

2.4 The budget constraint and the hours choice

$$c + p_v v + a' \;=\; y^n + a \;-\; p_h h'\,\mathbf{1}\{h'\neq h_1\} \;+\; Vch + F, \qquad c>0,\ v\geq 0,\ a'\geq \underline{a}. \tag{20}$$

Note what is absent: treatment, which costs utility, and bequests, which go to the government. Given a' and h', write R for the resources left over. Cobb-Douglas preferences over $(c, v+b)$ split them in closed form,

$$c = \frac{\alpha}{\alpha+\gamma}\big(R + p_v b_{dmh'}\big), \qquad v = \max\Big\{0,\ \frac{\gamma}{(\alpha+\gamma)p_v}\big(R + p_v b_{dmh'}\big) - b_{dmh'}\Big\}, \tag{21}$$

so the inner allocation never needs a numerical search, and the corner in v is where the household abstains because the Stone-Geary endowment already exceeds its desired intake. That is how a large b deters use, and it is the same mechanism as the threshold in Section 1.2.4.

Hours solve

$$u_c(c)\,\frac{\partial R}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}. \tag{22}$$

The clawbacks in (19), at a rate of one for food stamps and κ_r for the voucher, make $\partial R/\partial n$ jump *up* at the hours where each transfer is exhausted, because the clawback stops. The residual is therefore not monotone in n and (22) can have several roots, so it is solved by comparing the candidate roots rather than by a first order condition alone. This is the welfare trap, and it is the channel through which conditionality changes labour supply rather than only consumption.

2.5 Housing, and the homelessness loop

Housing is a four point grid. h_1 is literal homelessness, sleeping rough or in an emergency shelter, which is the HUD point in time definition and excludes doubling up; it carries a small service flow and no rent. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{23}$$

so two parameters describe the grid. Rent p_h is normalised to one and the *grid* is the calibrated object, since rescaling the grid by λ and the price by $1/\lambda$ leaves every budget constraint untouched and identifying the level would run through a weak channel.

Homelessness then enters four separate equations: survival (27), the employment transitions (28), the recovery hazard (25) and the Stone-Geary shifter (16). It raises mortality, worsens both employment margins, lowers the chance of recovering, and raises the marginal utility of the drug. Those four together are what make homelessness and the disorder mutually reinforcing rather than merely correlated, and it is that loop, absent from Section 1.2, that a housing programme acts on.

2.6 Drug use, the disorder, recovery and treatment

A household either abstains or uses, with v chosen continuously by (21). A clean household that uses acquires the disorder with the hazard of equation (2),

$$\Pr(d' = 1 \mid d = 0, v) = \pi_d(v) = 1 - e^{-\lambda v}, \tag{24}$$

a single hazard rising in the quantity used, with no light and heavy split and no threshold, so as in Section 1.2 the intensive and extensive margins are not separate decisions. A household with the disorder recovers with probability

$$\pi_r(h', \tau, m) = \min\left\{1,\ \bar{\pi}_r\, \mu_H^{\mathbf{1}\{h'=h_1\}}\, \mu_m^{\mathbf{1}\{m=1\}}\, g^{\tau}\right\}, \tag{25}$$

with $\mu_H < 1$, recovery is harder on the street, $\mu_m < 1$, harder with mental illness, and $g > 1$ the treatment effect.

Treatment τ is a *choice*, not a mechanical hazard, taken at the utility cost $\bar{\tau}^u$ in (15). This matters more than it looks: the policy under evaluation is called Treatment First, and g is the parameter through which its entire effect runs, so treatment has to be something a household decides to undertake and can decline. Section 1.5, where the only margin is whether to use, has no counterpart for it.

2.7 Serious mental illness

m is an exogenous two state Markov chain representing *serious* mental illness, about 5.5 per cent of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and includes mild anxiety: a state that halves productivity cannot sensibly cover a quarter of the population. The onset rate is derived from the target prevalence p_m and the recovery rate λ_m,

$$\text{onset}_m = \lambda_m \frac{p_m}{1 - p_m}, \qquad \Pr(m' = 1 \mid m = 0, d) = \text{onset}_m \cdot \iota^{\mathbf{1}\{d=1\}}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m, \tag{26}$$

so calibrating persistence never disturbs prevalence, and $\iota > 1$ makes the disorder raise the hazard of mental illness. That last channel runs in the direction the evidence supports: substance induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near a quarter, and in the Danish registers any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, opioid use disorder 2.74.

Mental illness then affects six objects: productivity conditional on working through ρ_m in (17), both employment margins through κ_{uem}, mortality through $\kappa_{\phi m}$, the recovery hazard through μ_m, the Stone-Geary shifter through ρ_m in (16), and eligibility for disability income through ζ_{DI}.

The mental health chain does *not* depend on housing, and that is a result rather than an omission. In the At Home / Chez Soi randomised trial, 1,198 homeless adults with mental illness across four cities, assignment to housing with intensive case management produced no detectable effect on symptom severity at any wave, while producing large effects on housing stability, its primary outcome. Making the channel housing dependent would mean adding a mechanism the best available trial looked for and did not find.

2.8 What the disorder and homelessness cost

Survival is

$$\phi_j(h', d, m) = 1 - \min\left\{1,\ (1 - \bar{\phi}_j)\,\kappa_{\phi H}^{\mathbf{1}\{h'=h_1\}}\,\kappa_{\phi d}^{\mathbf{1}\{d=1\}}\,\kappa_{\phi m}^{\mathbf{1}\{m=1\}}\right\}, \tag{27}$$

with $\bar{\phi}_j$ the baseline age profile and the multipliers themselves varying with age, rather than the single κ of (3). Because $\aleph > 0$ a higher death hazard is a real cost, and it is larger for households with more life remaining and more to consume, which is precisely the sorting Section 1.4 isolates.

Job finding and separation are

$$\pi^{ue} = \min\left\{1,\ \bar{\pi}^{ue}\,\kappa_{ueH}^{\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{\mathbf{1}\{d=1\}}\kappa_{uem}^{\mathbf{1}\{m=1\}}\right\}, \qquad \pi^{eu} = \min\left\{1,\ \bar{\pi}^{eu}\,\kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}}\kappa_{ued}^{-\mathbf{1}\{d=1\}}\kappa_{uem}^{-\mathbf{1}\{m=1\}}\right\}, \tag{28}$$

so each multiplier below one makes work simultaneously harder to find and easier to lose. In the baseline there is no direct wage penalty conditional on working, $\rho_d = 1$: the robust findings are on the employment margin, roughly half again as much unscheduled absence and 44 per cent higher turnover, while the wage penalty conditional on working is contested. A version calibrated to the income ratio of those with the disorder is reported as an alternative.

2.9 The voucher, and the two policies

A household without a voucher draws one with probability π_V if it is income eligible, and a household with one keeps it while eligible, so unlike Section 1.2.3 the voucher is persistent rather than redrawn each period. Eligibility is

$$y^g + \max\{0, ra\} \ < \ \kappa_V \cdot \text{median income}. \tag{29}$$

Under Housing First that is the whole rule. Under Treatment First the voucher is additionally lost with probability p_{det} by a household that used and with probability p_{fp} by one that did not,

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{HF} \times \begin{cases} 1 - p_{\text{det}} & \text{if } v > 0, \\ 1 - p_{\text{fp}} & \text{if } v = 0. \end{cases} \tag{30}$$

This is the one place where the quantitative model changes the *logic* of Section 1.5 rather than its setting. There, conditionality is perfectly enforced and perfectly met, so Treatment First deters everyone and nobody is thrown off the rolls. Here detection is imperfect and so are households: some fail the condition because they cannot get clean, and some fail it because the test is wrong. Those two groups are the empirical case for Housing First, neither exists in the stylized model, and that is why its welfare numbers are illustration.

2.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn from five independent transitions. Collecting them in one place:

Productivity y. Fixed over the life cycle. It carries the cross sectional dispersion of permanent income and the intergenerational link, and it is a type rather than a process, so $y' = y$. This is a departure from the submitted paper, where y followed a persistent autoregression, and it is made for two reasons: it is what keeps the child's penalty (37) a shift along a single scalar rather than a shift across the nodes of an age varying discretisation, and unemployment is the income risk that links the disorder to housing, which is retained in full. The stochastic version is the robustness direction.

Employment e. On separation the household draws insurance with probability π_{UI}, so with π^{ue} and π^{eu} from (28),

$$e = 1 \ \longrightarrow \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu}\pi_{UI} \\ e' = 3 & \text{w.p. } \pi^{eu}(1 - \pi_{UI}) \end{cases} \qquad e \in \{2, 3\} \ \longrightarrow \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}, \end{cases} \tag{31}$$

which is where insurance lasting exactly one year comes from: state 2 is never entered from state 2.

The disorder d. By (24) if $d = 0$ and (25) if $d = 1$, conditioning on the chosen v, h' and τ.

Mental illness m. By (26), conditioning on d.

The voucher η. By (29) and (30), conditioning on next period's eligibility and, under Treatment First, on the chosen v.

Write $\Gamma\big(s' \mid s,\, n, h', v, \tau\big)$ for the product of these five, which is the object the expectation in (32) is taken against.

2.11 The household's dynamic problem

Putting it together, for $j = 1, \dots, J$ and with V_{J+1} given below,

$$V_j(a, y, e, d, m, \eta) = \max_{n,\, h',\, v,\, \tau,\, a'} \left\{ \frac{\big[c^{\alpha}(h')^{\beta}(v + b_{dmh'})^{\gamma}\big]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x\big(H + \omega_x H_D\big) \right.$$
$$\left. +\ \beta\,\phi_j(h', d, m)\int V_{j+1}(s')\,\Gamma\big(\mathrm{d}s' \mid s, n, h', v, \tau\big) \right\} \tag{32}$$

subject to the budget (20), the income definitions (17) to (19), the choice set (14), $\tau = 0$ whenever $d = 0$, $n = 0$ whenever $j \geq j_R$ or $e \neq 1$, and $a' \geq 0$. The terminal condition carries the altruism,

$$V_{J+1} \;=\; \omega\,\mathbb{E}\,V_1\big(0,\, y_{\text{child}},\, 1, 0, 0, 0\big), \tag{33}$$

with y_{child} drawn from (37). At $\omega = 0$ death is worth zero and the damage a household does to its child is entirely external; at $\omega > 0$ the household internalises a share ω of it, and only $1 - \omega$ remains external. Since the child's productivity depends on whether the parent carried the disorder or was homeless, and both depend on v and h', equation (33) is what puts the child inside the parent's drug and housing decision.

The problem is solved by exact backward induction. Within a housing choice the split of c and v is closed form by (21) and hours come from (22), so the only search is over a', and it is done on the full grid rather than by a coarse then refine heuristic, because the objective is *not* single peaked in a': eligibility (29) is a step function of income, which includes ra; the food stamp floor kinks; and housing is discrete.

2.12 The government

The government pays social security, insurance, disability, food stamps and vouchers, and bears a fiscal cost χ_H per homeless household and χ_D per household with the disorder, including the cost of the drug tests that make detection possible. It finances this with the two taxes in (18) and the estates of the deceased, and the level parameter τ_{ℓ} clears the budget,

$$\underbrace{\textstyle\sum y^{g} - \tau_{\ell}\sum (y^{g})^{\xi} + \tau_k \sum \max\{0, ra\} + \text{estates}}_{\text{revenue}} \;=\; \underbrace{SS + UI + DI + F + Vch + \chi_H H + \chi_D D + G}_{\text{outlay}}. \tag{34}$$

Estates accrue to the government rather than being redistributed as a lump sum, because a uniform transfer to the poor has welfare consequences of its own that would contaminate every policy comparison in the paper, and because it removes an equilibrium fixed point.

Equation (34) is the single most important thing the stylized model does not have. There, the two policies happen to cost the same and the comparison is unaffected. Here a policy that houses more households spends more, τ_{ℓ} rises to pay for it, and the higher tax distorts (22) and saving for everyone, including the households the programme never reaches. That distortion has no counterpart in Section 1.6, it runs against Housing First, and it is the main reason the dominance result (11) does not survive into the quantitative model intact.

2.13 The drug market

Drug production has cost $C(V) = \frac{\psi_1}{\psi_2} V^{\psi_2}$, so supply satisfies

$$p_v = \psi_1 V^{\psi_2 - 1}, \qquad V = \int v(s)\, \mathrm{d}\Phi(s), \tag{35}$$

with price elasticity of supply $1/(\psi_2 - 1)$ and ψ_1 normalised so that $p_v = 1$ at the benchmark quantity. Endogenising the price does not suppress the normalisation of Section 1.2.3, it moves it to ψ_1.

The baseline is a flat supply curve, which is the maintained assumption in the drug economics literature and has a natural experiment behind it. In 2010 the reformulation of OxyContin pushed a very large mass of demand into heroin, enough to quadruple heroin death rates, and purity adjusted heroin prices did not rise; they fell slightly. A demand shift of that size leaving price flat is direct evidence of a near horizontal supply curve, so finite elasticities are the robustness direction rather than the baseline.

2.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \phi_j \int \Gamma\big(\mathcal{S}' \mid s,\, n_j(s), h'_j(s), v_j(s), \tau_j(s)\big)\, \mathrm{d}\Phi_j(s) \qquad \text{for every measurable } \mathcal{S}', \tag{36}$$

where ϕ_j is evaluated at the household's own $(h'_j(s), d, m)$. The mass that does not survive, plus the mass at $j = J$, is replaced by newborns at $j = 1$ with zero assets, employed, clean, without mental illness and without a voucher, whose productivity is drawn from the unconditional distribution Ω_y scaled by the penalty,

$$y_{\text{child}} \;\sim\; \Omega_y \;\text{ scaled by }\; (1-\delta_d)^{\mathbf{1}\{\text{parent had the disorder}\}}\,(1-\delta_h)^{\mathbf{1}\{\text{parent was homeless}\}}. \tag{37}$$

This is equation (12) made operative, and two implementation points matter. The penalty shifts the newborn's draw *probabilities* over the existing productivity nodes rather than creating penalised types, which would multiply the productivity space by four for no gain. And the parent's status is read at death, which understates exposure, since a parent who acquired the disorder at forty and recovered at sixty leaves no mark; the resulting externality is therefore a lower bound.

2.15 Stationary equilibrium

A stationary equilibrium is a set of value functions $\{V_j\}$, policy functions $\{n_j, h'_j, v_j, \tau_j, a'_j\}$, a distribution Φ, and prices and aggregates $\big(\tau_\ell,\, p_v,\, \bar{Y},\, \text{median},\, H,\, H_D\big)$ such that

1. the policy functions solve (32) given the terminal condition (33), and $\{V_j\}$ are the associated values;
2. Φ is invariant under the law of motion (36) together with the newborn rule (37), and the mass of newborns equals the mass of the deceased;
3. the government budget (34) holds at τ_ℓ;
4. the drug market clears at p_v, so (35) holds with $V = \int v(s)\, \mathrm{d}\Phi$;
5. $\bar{Y}$, the median of gross income, the homelessness rate H and the homeless share carrying the disorder H_D are the aggregates implied by Φ.

Three of those are genuine fixed points rather than accounting. H and H_D enter (15), so whenever $\kappa_x > 0$ the homelessness rate is an argument of the utility function that the distribution itself determines. The median enters eligibility (29), and in a skewed income distribution it sits well below the mean, so using the mean would materially overstate the eligible population. And τ_ℓ is what makes the policy comparison a general equilibrium object at all: the two policies do not cost the same once take up and detection are imperfect, so they do not face the same tax, and the household problem is different under each for reasons that have nothing to do with the voucher rule itself.

2.16 What is left out, and what it would do

The disamenity κ_x is priced and set to zero in the benchmark. Loubière et al. (2020) ask adults in eight European countries what they would pay in annual taxes to end homelessness through Housing First: the mean over all respondents is €23 a year and €57 among the 42 per cent willing to pay anything, which is 0.08 and 0.20 per cent of average annual earnings. That is small enough that it cannot move a ranking on its own, so the benchmark sets $\kappa_x = 0$ and the ranking is reported across a range, where it can only strengthen the case for whichever policy houses more people. Zero is a defensible baseline here in a way it would not be for a parameter the conclusion turns on: it simply means no externality, so the reported results are a lower bound on the case for housing.

3 Calibrating the quantitative model

Two stages, and they are different exercises. The first matches a cross section of the stationary equilibrium, nineteen parameters against twenty one moments. The second matches short run *responses* measured in experiments, which cannot be read off a comparison of steady states. Only what is new or what Section 1.3 does not already illustrate is described here.

The same cliffs, and the same answer. Three moments are step like and are bisected rather than stepped, for the reason Section 1.3 gives: participation is a near threshold function of b, homelessness moves from 0.12 to 3.29 per cent for a 35 per cent change in h_{HL}, and wealth over output is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves. The parameter to moment map is close to triangular and the sign of $\partial M/\partial\theta$ is declared for every pair, so a mistaken sign shows up as a moment walking away from its target rather than as a silent failure.

Anchors. The discount factor targets aggregate wealth at three times output, with output measured as all factor income and transfers excluded, which is the standard US private capital to output ratio rather than the SCF's net worth to household income ratio of about 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold. As in Section 1.3, ℵ is set to a value of a statistical life rather than to a behavioural moment, because a behavioural target pins ℵ's effect on *who* uses without pinning its level, and at the wrong level households prefer death and the mortality cost of drug use becomes a benefit.

Three targets that were wrong and are now right. Substance use disorder among the homeless is 21.7 per cent, the *drug* use disorder figure from Gutwinski et al.'s pooling of thirty nine studies, not the 36.7 per cent that is alcohol. The income of drug users is split: 95.1 per cent of average for users and 68.7 per cent for moderate or severe disorder, ages 26 and over, with the disorder ratio targeted and the use ratio kept for validation, which makes the economics explicit, since use is close to income neutral and it is progression to the disorder that concentrates at the bottom. The housing burden is targeted over all households rather than over renters only.

Second stage: responses, not steady states. These are partial equilibrium cohort experiments. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at benchmark; the household problem is solved again, because each arm changes the rules a forward looking household faces. Milby et al. (2005) randomised 196 cocaine dependent homeless adults into three arms on identical day treatment, and the targeted moment is the ratio of the unconditional voucher arm to no assistance, $6.60/4.48 = 1.473$ weeks abstinent. The conditional against unconditional comparison is deliberately *not* targeted, because the source reports it as insignificant in the raw randomised comparison; calibrating to it would repeat the error this revision is correcting elsewhere. The drug price participation elasticity, -0.63, is the second stage's other target.

What the two new parameters are matched to. The weight on the child ω and the transmission penalty δ_d are fitted *together*, because each moves the other's moment. Altruistic parents use less, so the

share of children born to a parent with the disorder falls, so the transmission ratio at a given δ_d falls with ω.

Parameter	Moment	Source
ω	Drug use falls 15.5 per cent with a child present	NSDUH 2024, ages 26+, parents reweighted to the non parents' joint age, income, sex and marital distribution
δ_d	Child's disorder rate relative to a child of a clean parent	The epidemiological range of two to four
δ_h	Set, not calibrated	Chetty, Hendren and Katz (2016)

Since every household in the quantitative model has a child, there is no parent and non parent comparison to run inside it. The condition is the counterfactual instead: ω is the entire reason a household uses less than it otherwise would, so switching altruism off at the calibrated parameters must raise drug use by 18.4 per cent. Run through the stylized model that condition gives $\omega = 0.014$, which is worth stating because it is small: the ranking in Section 1.6.2 reverses back to Housing First only near $\omega = 0.43$, so the calibrated weight sits a factor of thirty inside the region where the externality still decides the comparison, and it is consistent with Altonji, Hayashi and Kotlikoff (1997), who reject altruism operating at the margin at all. The measured gap is an upper bound, since users also lose custody of children, so the ranking is reported across zero to that value.

Tables 5 and 6 list every parameter of the model: the thirty nine that are assigned, whether by normalisation, from the literature, directly from data, in equilibrium or as a reported sensitivity, and the twenty two that are calibrated inside the model, each against the moment it is assigned to.

Table 5: Assigned parameters of the quantitative model

Symbol	Meaning	Value, source or target
σ	risk aversion	2.0. Attanasio and Weber (2010) survey; the standard life cycle value
σ_ℓ	curvature on leisure	3.0; Frisch elasticity 0.5, Chetty (2012)
ρ_H	b multiplier, homelessness	$\rightarrow$ housed/unhoused abstinence ratio, 1.473. Milby et al. (2005), AJPH 95(7):1259 to 65, NACH versus NH arms
p_v	drug price	clears the drug market; = 1 at the benchmark by the normalisation of ψ_1
r	risk-free rate	0.03, Kaplan, Mitman and Violante (2020)
κ_x	externality weight	Inverted from Loubière et al.'s eight-country contingent valuation, €23 per adult per year to end homelessness, normalised per unit of homelessness by the European reference rate. This is the mean over ALL respondents. The €57 quoted in an earlier draft is the mean among the 42.3%...
ω_x	extra weight on homeless-with-SUD	1.0 in the baseline: the disamenity depends on the headcount alone. No direct estimate exists, R2's wording ("especially if many are addicted to drugs") implies a positive weight and the hedonic split between supportive housing and emergency shelter implies composition matters, but neither...
ξ	tax progressivity (HSV)	0.85, Heathcote, Storesletten and Violante (2017)
τ_ℓ	tax level	clears the government budget the budget given the distribution
τ_k	capital income tax	0.15. US long run capital-gains and qualified-dividend rate for the middle brackets
κ_V	voucher eligibility threshold	0.50 of median income, HUD rule
κ_r	tenant contribution rate	0.30, HUD rule
p_{test}	probability a voucher holder is tested, TF	1.0 at the benchmark, swept downward. Detection now requires a test, so effective detection is $p_{\text{test}}p_{\text{det}}$ and the old behaviour is the $p_{\text{test}} = 1$ corner
c_{test}	cost per voucher holder tested	0.0023 of average income. A urine panel runs about $40 and quarterly testing is roughly $160 a year against average earnings near $70,000. It enters the GOVERNMENT BUDGET, where it competes with everything else the government pays for, R1's comment 4(c) is that the $1 to 2bn estimate is too...
p_{det}	detection probability given a test	1.0 in the benchmark, swept downward. No estimate exists and none should: R1 4(a) and AE A11 object precisely to assuming perfect enforcement, so the point is the sweep
p_{fp}	false-positive rate under TF	0.0 in the benchmark, swept upward. As above, R1 4(b)

Symbol	Meaning	Value, source or target
χ_H	fiscal cost per homeless household	0.4286. The homeless state h_1 is literal homelessness, shelters and the street, not doubling up, so the matching figure in Abramson and Van Nieuwerburgh (2024, NBER w32582, p. 30) is their $2,500 per household per month for the literally homeless, not the $1,775 they blend across...
χ_D	fiscal cost per household with SUD	0.212; CEA (2017), ≈$30,000 per person-year
G	other public spending	Residual that balances the budget at the benchmark, then held fixed across counterfactuals so policy is financed by τ_ℓ rather than by cutting other spending
ρ_y, σ_y	AR(1) in log productivity	0.90, 0.20. Storesletten, Telmer and Yaron (2004); Floden and Lindé (2001)
ε_j	deterministic age profile	quadratic in age, fitted to the PSID/CPS hump. Replace with a directly estimated profile
$\bar{\phi}_j$	baseline survival	Done. SSA period life table (oact/STATS/table4c6), ages 26 to 85, the two sexes combined by survivor weight $q(x) = [l^M q^M + l^F q^F]/[l^M + l^F]$ rather than averaged equally, by the mid-eighties the survivor populations differ enough that equal weights are simply wrong. Loaded from...
$\bar{\pi}^{ue}$	job finding, annual	0.55, Krueger, Cramer and Cho (2014)
π_{UI}	share of entrants drawing UI	0.30; insured unemployment / total unemployment, FRED 2014 to 24
ϱ_{UI}	UI replacement rate	0.50. DOL statutory replacement rates average close to half of prior weekly wages
κ_{ueH}	employment multiplier, homeless	0.80, Desmond and Gershenson (2016)
$\kappa_{\phi H}$	mortality multiplier, homeless	Now varies with age: 4.41 at 26, 3.80 at 44, 3.14 at 56, 1.09 at 80. Baggett et al. (2013) give race adjusted all cause mortality rate ratios for homeless adults of 8.6 (men) and 9.6 (women) at 25 to 44, 4.5 at 45 to 64 and 1.1 at 65 to 84. The previous flat 1.60 was wrong twice: five times...
$\kappa_{\phi d}$	mortality multiplier, SUD	Shape from data, level still calibrated. → drug price participation elasticity, −0.63 (inside the −0.36 to −0.55 cocaine and −0.80 to −0.90 heroin participation range of Saffer and Chaloupka), bracketed by the 2.5 to 9.8 SMR range of Degenhardt et al. (2011). The age...
$\kappa_{\phi m}$	mortality multiplier, SMI	2.50, and constant in age, which is a finding rather than a default. Unlike SUD and homelessness, whose multipliers collapse with age, Olfson et al. (2015) give all cause SMRs for schizophrenia of 3.6 at 20 to 34, 3.4 at 35 to 54 and 4.0 at 55 to 64: flat, if anything rising. The level is a...
ρ_d	SUD productivity penalty	1.00 in the baseline: no direct wage effect. The employment-margin evidence is robust (about 50% more unscheduled absence, 44% higher turnover) and is carried by κ_{ued}; the wage penalty conditional on working is contested, with parts of the older literature finding positive effects...
j_E	maximum experimentation age	Real age 55, with no source, R1's smaller comment 1 says so. Run with and without (Section the text); if it can be dropped it should be
$\bar{\pi}_r$	baseline recovery hazard, housed	0.025/yr; Vaillant (1995). The AE questions applying alcohol recovery rates to drug SUD; Fleury et al. (2016) give 35 to 54% remission at a mean 17-year follow up across substances, which brackets this
μ_m	recovery multiplier, SMI	0.82. Dual-diagnosis cohorts relapse at 39.8% against 26.4% without a co-occurring disorder (odds ratio 1.85 for difficulty maintaining abstinence), so sustained remission runs 0.602 against 0.736, a ratio of 0.82; converting the odds ratio to a risk ratio at that base rate gives the same...
g	treatment effect on recovery	1.35, from an odds ratio of 1.22 for contingency management in the only clean randomised evidence. Specialty-treatment remission of 47.6% and overall remission of 35 to 54% are not treatment effects, different populations, different selection, which is why they are not used here. The TF...
λ_m	SMI recovery rate	0.067/yr; İmrohoroğlu, Zhao's own moment, ≈50% improve enough to work within 10 years
ι	SUD multiplier on SMI onset	2.0, new. Danish registers: any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, amphetamine use disorder 1.90, opioid use disorder 2.74; substance-induced psychosis converts at a pooled 25% (Murrie et al.). Range 1.3 to 2.7
$\bar{y}_{DI}$	disability benefit level	0.25 of average income. SSA: average SSDI benefit $1,537 a month in 2024, about $18,400 a year against average annual earnings near $70,000, so 0.26. The SSI federal benefit rate of $943 a month is 0.16; SSDI is the relevant figure for people with a work history
ψ_1	drug supply shifter	Normalised so $p_v = 1$ at the benchmark quantity, which is what lets the price be endogenous without disturbing the existing normalisation (R3 M2d)

Symbol	Meaning	Value, source or target
ψ_2	drug supply curvature	Elasticity $1/(\psi_2 - 1)$, baseline 2.0, bracketed from measured tax incidence in two legal markets. Lower bound 0.79: Hansen, Miller and Weber (2018) on Washington's 2015 switch from a 25% gross receipts tax at every supply chain stage to a single 37% retail excise, in universal state...

Table 6: Internally calibrated parameters of the quantitative model

Symbol	Meaning	Value, source or target
ω_ℓ	weight on leisure	→ average hours worked, 40% of time. Guvenen et al. (2023)
θ_h	Cobb–Douglas weight on housing	→ aggregate housing cost / income, 21.2%. SCF 2022, ages 26 to 85, all households, owner housing at imputed rent; data/scf_housing_burden.py
θ_v	Cobb–Douglas weight on drugs	→ drug spending / income given SUD, 44%. Carried over from the submitted paper; its construction must be stated explicitly in the appendix (R1 2e)
b_0	Stone–Geary shifter, clean	→ drug participation, 5.13%. NSDUH 2024 microdata, ages 26+; data/nsduh_gradients.py
b_1	Stone–Geary shifter, SUD	→ abstinence rate among homeless adults with SUD, 18.7%. Milby et al. (2005), NO-HOUSING arm: 4.48 consecutive weeks abstinent out of 24. b_1 governs how costly abstention is once a household already has SUD, so one arm of the trial pins the level and the contrast between arms...
ρ_m	b multiplier, mental illness	→ SUD rate given SMI, 9.75%. NSDUH 2023 detailed tables, ages 26+, by Bayes on Tables 6.13B, 6.4B and 5.10B
$\aleph$	flow value of being alive	→ model implied VSL, \$7.5m (Viscusi and Aldy 2003; EPA 2010). It used to target the SUD income ratio, which pins $\aleph$'s effect on WHO uses but not its LEVEL, and at the resulting level 64.6% of the population had $V < 0$, i.e. preferred death, which turns the mortality cost...
$\bar{\tau}^u$	utility cost of treatment	→ treatment take up given SUD, ≈24%. NSDUH 2023: about a quarter of people with a past-year SUD received any substance use treatment. Verify against the detailed tables
β	discount factor	→ wealth / GDP = 3. BEA fixed assets against GDP; the standard US private-capital-to-output ratio
h_{HL}	service flow of homelessness	→ homelessness rate, 0.399%. HUD PIT 2023, converted to households: 467,020 homeless individuals, one household each, plus 57,563 homeless family households, against 131,332,360 occupied units. The model's agent is a household, so the person rate of 0.195% is the wrong...
$\underline{h}$	smallest rental size	→ share with housing cost > 50% of income, 17.49%. SCF 2022; data/scf_housing_burden.py
ς	geometric spread of rental sizes	→ share with housing cost > 30% of income, 40.62% (SCF 2022). Data now in hand for the better target: ACS B25042 (Tenure by Bedrooms, occupied units, 2023, $N = 131{,}332{,}360$) gives 13.21% / 24.12% / 62.67% across $\{0, 1\}$, $\{2\}$, $\{3+\}$ bedrooms. All households, not...
$\underline{c}$	food-stamp consumption floor	→ food stamps / labour income, 0.42%. USDA Food and Nutrition Service, SNAP outlays FY2023, against GDP
π_V	voucher lottery probability	→ voucher recipients / income-eligible, 26.6%. HUD Picture of Subsidized Households against the income-eligible renter population
$\bar{\pi}^{eu}$	job separation	→ unemployment rate, 4.9%. BLS, ages 25+, average over the recent expansion
κ_{ued}	employment multiplier, SUD	→ unemployment rate given SUD, 16.25%. Corrected, and the previous figure's stated source was wrong. The note attributed 26.9% to a tabulation in data/nsduh_gradients.py; that script contains no employment variable and cannot have produced it. Measured now from the...
κ_{uem}	employment multiplier, SMI	→ unemployment rate given SMI, 7.1%. NSDUH Table 6.5B: SMI prevalence 8.9% among the unemployed against 5.1% among the full-time employed, inverted by Bayes
ρ_m	SMI productivity penalty	→ homeless income / average, 19.1%. Meyer et al. (2025), homeless population linked to administrative earnings records
π_d	addiction hazard given any use	→ SUD rate, 3.68%. NSDUH 2024 microdata, ages 26+
μ_H	recovery multiplier, homeless	→ drug use disorder share of the homeless, 21.7%. Gutwinski et al. (2021), PLOS Medicine, 39 pooled studies. Note this is the DRUG figure; the submitted paper's 37% is their ALCOHOL figure
p_m	SMI prevalence	→ 5.5%, NSDUH 2023, adults. A target rather than a parameter now, since onset varies with SUD

Symbol	Meaning	Value, source or target
ζ_{DI}	disability coverage given SMI	$\rightarrow$ SMI share of the homeless, 25%. Gutwinski et al. (2021): 12.4% schizophrenia-spectrum plus 12.6% major depression. This is İmrohoroğlu, Zhao's own identifying margin (ζ_{DI} in their notation)

What fixing permanent income costs, and what has to be refitted. The quantitative model holds permanent productivity fixed over the life cycle and keeps only the unemployment risk, which is what makes the child's penalty a shift along a single scalar rather than a shift across the nodes of an age varying discretisation. Two things then have to be redone rather than inherited. The dispersion of log productivity has to carry on its own what the fixed effect and the shock carried together, so it is set to the average cross sectional variance the stochastic process delivered rather than to its value at the youngest age. And the wealth moments are the ones at risk, since removing income risk removes precautionary saving: β is refitted to wealth over output, and the share of wealth held at the bottom of the distribution becomes an untargeted moment worth reporting rather than a target quietly still being hit.

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